

P is projective if for any $M \xrightarrow{\varphi} N \rightarrow 0$

we have

$$\begin{array}{ccc} & \exists h & \downarrow P \\ & \swarrow & \downarrow \\ M & \xrightarrow{\varphi} & N \rightarrow 0 \end{array}$$

: if $\varphi: M \rightarrow N$ is surjection and P is projective then $\text{Hom}(P, M) \rightarrow \text{Hom}(P, N)$ is surjective.

$$P(i) : \in Q_0 \quad P(i) = ((P(i))_j, \varphi_2)$$

where $P(i)_j$ has basis which consist of all paths from i to j

$$i \xrightarrow{\alpha_1} \dots \xrightarrow{\alpha_r} j \quad c = (i | \alpha_1, \alpha_2, \dots, \alpha_r | j)$$

for Q we need it has no oriented cycles

$$\sum \lambda_c c$$

where c runs through paths from i to j

$$\lambda_c \in k$$

$$\text{let } \alpha = j \rightarrow i$$

$$\varphi_2 : (P(i))_j \rightarrow (P(i))_i$$

$$\sum \lambda_c c \mapsto \sum \lambda_c \varphi_2(c)$$

c paths from i to j

$$c : (i | \alpha_1, \dots, \alpha_r | j) \mapsto \varphi_2(c) = (i | \alpha_1, \dots, \alpha_r, \alpha | i)$$

we have shown that $P(i)$ is projective

$$(P(i))_i = k \quad c = (i | i)$$

analogously. $I(i)$ is

$$S(i) \quad S(i)_j = \delta_{ij} k = \begin{cases} k & j=i \\ 0 & \text{otherwise} \end{cases} \quad \alpha: j \rightarrow i \quad \varphi_2 = 0$$

$S(i)$ are simple, and these are only simple representations in $\text{Rep } Q$

proof: $S(i)$ simple because of $\dim S(i) = (0, 0, \dots, 1, 0, \dots, 0)$

Let M be any representation in $\text{Rep } Q$
 then since Q does not contain oriented cycles
 it contains sink

a vertex $i \in Q_0$ is called a sink if there is no arrow $\alpha: i \rightarrow j$
 $\{ \omega \} \neq i$ for $\forall \alpha \in Q_1$.
 a vertex $j \in Q_0$ is called a source if there is no arrow $\beta: i \rightarrow j$
 $\{ \beta \} \neq j$ for $\forall \beta \in Q_1$.

$i \in Q_0$ is a sink $\Leftrightarrow p(i) = s(i)$ is simple

$j \in Q_0$ is a source $\Leftrightarrow I(j) = \{ \omega \}$ is simple

Let $M = (M_i, \varphi_\alpha)$

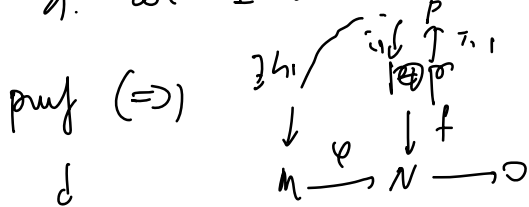
We want to find in M such vertex i with $M_i \neq 0$
 and $m_{ij} = 0$ for any $\alpha: i \rightarrow j$
 and $m_{ij} = 0$ for all $\alpha: i \rightarrow j$.

then $s(i)$ is subrep of $M \Rightarrow s(i)$ are all the simple modules.

prop: 1) let P and P' be representations of Q , P, P' are proj. $\Leftrightarrow$

$\text{Rep } Q$ is proj.

2) let I and I' be in $\text{rep } Q$ I, I' inj $\Leftrightarrow I \oplus I'$ inj



$$\exists h_1 \text{ s.t. } \varphi \circ h_1 = f \circ i.$$

$$\text{def } h: P \oplus P' \rightarrow M \text{ s.t. } h = h_1 \circ \pi_1$$

is inj.

prop. $s(i)$ $p(i)$ $I(i)$ are indecomposable for any $i \in Q_0$

Thm $M \in \text{rep } Q$, then $\text{Hom}(P(i), M) \cong M_i$ for any $i \in Q_0$

proof. Let $f: P(i) \rightarrow M$. $\psi(f) = f_i(i|i)$
 $f_i(P(i)_j) \rightarrow M_j$ $(P(i))_i = (e(i))$

it is well-defined and is linear.

$$P(i) = (P(i)_j, \varphi_2)$$

$$M = (M_j, \beta_2)$$

1) let us show that ψ is injective.

$$\text{supp } \psi(f) = f_i(i|i) = 0$$

$$\text{for any } f_j: (P(i))_j \rightarrow M_j$$

$$(i = (i_1, i_2, \dots, i_r)_j) \quad f_j(i_1)$$

for any j with $\alpha: i \rightarrow j$

$$\begin{array}{ccc} (P(i))_i & \xrightarrow{f_i} & M_i \\ \psi_i \downarrow & & \downarrow \beta_i \\ (P(i))_j & \xrightarrow{f_j} & M_j \end{array}$$

$$\begin{array}{ccc} (P(i))_i & \xrightarrow{\quad} & M_i \\ \downarrow \psi & & \downarrow \beta \\ (P(i))_j & \xrightarrow{f_j} & M_j \end{array}$$

2) to show that ψ is surjective

choose $m_i \in M_i$, def $f \in \text{Hom}(P(i), M)$

$$\text{in the form } f: (P(i))_i \rightarrow M_i$$

$$(i|i) \rightarrow m_i \quad f_i(i|i) = m_i$$

then there $\psi(f) = m_i$, $\rightarrow$ define $j \neq i$ we follow path:

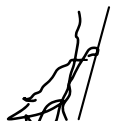
$$f_j: (P(i))_j \rightarrow M_j$$

for $i \in Q_0$ def $\psi(f_i)$

then

conclusion: def ψ is also

there are no proper maps



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